

**Cool summers are stormy summers: precession reorganizes extratropical
cyclone activity through the quiet season**

Xiaoxu Shi^a

^a *Alfred Wegener Institute, Helmholtz Centre for Polar and Marine Research, Bremerhaven,
Germany*

Corresponding author: Xiaoxu Shi, shixiaoxu@sml-zhuhai.cn

7 ABSTRACT: Extratropical cyclones cluster in winter, while precession reshapes the seasonal
8 insolation cycle most strongly in summer, so storm tracks might be expected to ignore the precession
9 cycle. This expectation is tested with twelve equilibrium simulations of the coupled climate
10 model AWI-ESM-2.1 that cover the precession cycle at intervals of thirty degrees under amplified
11 eccentricity and fixed obliquity, with extratropical cyclones tracked explicitly in six-hourly pressure
12 fields in both hemispheres. The annual-mean storm response is small. The seasonal response is
13 not. In the North Atlantic, the North Pacific, and the Southern Ocean the response concentrates in
14 local summer, where storm activity ranges across the phases by up to two thirds of its mean while
15 winter changes stay below about a tenth. We find that summers spent near perihelion are calm
16 and summers spent near aphelion are stormy, so midlatitude storminess in the two hemispheres
17 varies in antiphase across the precession cycle. The signal is carried by cyclone number and
18 position rather than by propagation speed, and it follows the pronounced summer response of the
19 Eady growth rate, which is dominated by vertical shear in all three basins. Month by month,
20 the response is phase locked to the calendar date of perihelion, delayed by one to two months of
21 surface thermal inertia. Precession therefore acts on extratropical storm tracks through their quiet
22 season. Summer-sensitive proxy archives should record a strong and hemispherically antiphased
23 storminess signal at the precession period that annually integrating archives would miss.

24 SIGNIFICANCE STATEMENT: Ice-age cycles are paced by slow changes in Earth's orbit, and
25 the strongest of these, precession, reshapes sunlight mainly in summer, while midlatitude storms
26 happen mainly in winter. Climate model simulations covering a full precession cycle show that
27 the orbit nevertheless controls storminess, by acting through summer. When perihelion falls in a
28 hemisphere's summer, that summer has fewer storms and the storm belt sits farther poleward, and
29 the two hemispheres respond in opposite phase. This tells researchers where to look for orbital
30 storm signals in geological records, namely in archives sensitive to summer, not to the yearly
31 average.

32 1. Introduction

33 Extratropical cyclones organize the weather of the middle latitudes. They carry most of the
34 poleward heat and moisture transport outside the tropics, deliver a large share of midlatitude
35 precipitation, and cluster into the storm tracks of the North Atlantic, the North Pacific, and the
36 Southern Ocean (Hoskins and Valdes 1990; Chang et al. 2002; Hoskins and Hodges 2002, 2005).
37 These storms grow by baroclinic instability on the meridional temperature gradient (Eady 1949),
38 and the gradient is steepest in winter. Cyclone numbers therefore peak in the cold season in both
39 hemispheres, in the south as clearly as in the north (Simmonds and Keay 2000). Extratropical
40 cyclones are creatures of winter.

41 Precession, in contrast, is a lever that acts on summer. The slow turning of the longitude of
42 perihelion is the strongest orbital pacemaker of the seasonal insolation cycle, yet it redistributes
43 sunlight rather than adding any, and the redistribution is proportional to the sunlight a season
44 already receives (Berger 1978; Laskar et al. 2004). Midlatitude summer insolation swings by more
45 than 100 W m^{-2} across a precession cycle while midwinter insolation barely moves, which is why
46 the orbital literature treats summer energy as the climatically active quantity (Huybers 2006). A
47 winter phenomenon facing a summer lever invites a simple expectation. Precession should have
48 little to say about storm tracks. That expectation has never been put to a clean test.

49 Where the imprint of precession has been studied in depth, it appears as a seasonal reorganization
50 of climate rather than a change in the annual mean. The monsoon literature carries this tradition,
51 from the classic orbital experiments of Kutzbach and Guetter (1986) to single-forcing designs that
52 isolate precession explicitly (Bosmans et al. 2015; Merlis et al. 2013), and cave records show

53 the northern and southern monsoons swinging in antiphase as perihelion migrates through the
54 year (Wang et al. 2008). The extratropics have received far less attention. Kaspar et al. (2007)
55 simulated a stronger and poleward-shifted North Atlantic winter storm track for the Eemian from
56 two orbital time slices, Brayshaw et al. (2010) traced Holocene orbital forcing into the North
57 Atlantic and Mediterranean winter storm tracks through the baroclinicity they support, and Varma
58 et al. (2012) found the Southern Hemisphere westerlies migrating under Holocene orbital forcing.
59 Explicit cyclone tracking has entered paleoclimate work mainly at the Last Glacial Maximum,
60 where ice sheets dominate the forcing (Pinto and Ludwig 2020; Raible et al. 2021), and in transient
61 simulations of the last millennium (Hess and Chemke 2025). For tropical cyclones the precessional
62 response has recently been isolated in high-resolution paleoclimate simulations (Raavi et al. 2023).
63 To our knowledge, no previous study has applied explicit cyclone tracking across a full precession
64 cycle to isolate the orbital response of the extratropical storm tracks.

65 Closing this gap matters beyond dynamics. However, the paleoclimate records that would register
66 past storminess, from dust and wind-blown sediment to westerly-belt precipitation, respond to
67 particular seasons rather than to the annual mean, and a seasonally biased archive can fabricate
68 or erase an orbital signal depending on which season carries it (Laepple et al. 2011). Knowing
69 where in the seasonal cycle the precessional storm signal lives is a prerequisite for reading such
70 records. Storm diagnostics raise a parallel issue, because measures of cyclone influence time,
71 cyclone counts, and cyclone intensity respond to forcing in different ways and must be separated
72 before any single proxy is interpreted.

73 In the present study the precession cycle is covered with twelve equilibrium simulations of a
74 coupled climate model, and every extratropical cyclone is tracked explicitly in six-hourly pressure
75 fields in both hemispheres. The design isolates precession at amplified eccentricity with all other
76 boundary conditions fixed, so that the seasonal anatomy of the orbital storm response can be
77 mapped basin by basin against the calendar month of perihelion.

78 The naive expectation fails, and it fails in an instructive way. The annual-mean storm response
79 is indeed small, but the signal has not vanished. It is concentrated in local summer, the quiet
80 season, in all three storm basins. Summers spent near perihelion are calm and summers spent near
81 aphelion are stormy, so the two hemispheres respond in antiphase across the precession cycle, an
82 extratropical counterpart of the monsoon seesaw. The response travels from insolation to storms

83 through the summer Eady growth rate, and the growth-rate change is carried by vertical shear
84 (Lindzen and Farrell 1980). The paper is organized around which season carries the precessional
85 storm response, what dynamical chain transmits it, and why the winter storm track stays orbitally
86 quiet. Section 2 describes the model, the ensemble, and the tracking. Section 3 presents the
87 seasonal anatomy of the response, and section 4 discusses mechanisms, implications for proxy
88 records, and limitations.

89 **2. Data and methods**

90 *a. Model and experiments*

91 The simulations are performed with AWI-ESM-2.1-wiso, an isotope-enabled version of the AWI
92 Earth System Model (Shi et al. 2023). The atmosphere component is ECHAM6 at T63 resolution
93 with 47 vertical levels (Stevens et al. 2013), which includes the land surface module JSBACH
94 (Raddatz et al. 2007). The ocean and sea ice component is FESOM2, a finite-volume model
95 formulated on an unstructured mesh (Danilov et al. 2017; Sidorenko et al. 2019). The water
96 isotope tracers carried by the model are not used in the present study.

97 Twelve equilibrium simulations are analyzed, taken from the idealized precession ensemble
98 introduced by Shi et al. (2025). The longitude of perihelion ω is prescribed at values from 0° to
99 330° in steps of 30° , while eccentricity is fixed at 0.058 and obliquity at 24.5° . The eccentricity
100 value sits near the upper bound reached during the Quaternary and amplifies the precessional
101 modulation of insolation, so that the response emerges clearly from internal variability. All
102 remaining boundary conditions follow the preindustrial setup, and each phase is branched from
103 a long preindustrial control. Thirty years of six-hourly output from the equilibrated part of each
104 simulation form the basis of the cyclone analysis. The calendar month of perihelion for each phase
105 is computed from two-body orbital geometry (Berger 1978), and the same formulary provides the
106 insolation fields shown in Fig. A3.

107 *b. Cyclone detection and tracking*

108 Extratropical cyclones are detected and tracked with the algorithm of Crawford et al. (2021),
109 and Fig. A4 summarizes the procedure with one six-hourly field of the ensemble. Six-hourly sea
110 level pressure is first interpolated onto 100-km equal-area grids centered on each pole, and the two

hemispheres are processed separately. A grid cell qualifies as a cyclone center when its pressure p_c lies below that of all eight neighbors and the surrounding field keeps a sufficient inward gradient. The gradient test compares p_c with the mean pressure $\bar{p}(d)$ on a one-cell ring at distance d from the candidate,

$$\frac{\bar{p}(d) - p_c}{d} \geq \gamma, \quad d = 1000 \text{ km}, \quad \gamma = 750 \text{ Pa (1000 km)}^{-1}, \quad (1)$$

and candidates over terrain above 1500 m are discarded. Centers closer than 1000 km are treated as one multicenter system. The cyclone area A is obtained by growing closed isobars around each center at a 200-Pa interval until a contour fails to close, and the last closed isobar defines the edge pressure p_e (Fig. A4b). Each cyclone carries the depth and the area-equivalent radius

$$D = p_e - p_c, \quad R = \sqrt{A/\pi}. \quad (2)$$

Tracking links the centers of consecutive six-hourly fields (Fig. A4c). For every active cyclone a first-guess position is projected from its previous displacement, damped to three quarters,

$$\mathbf{x}_q = \mathbf{x}_t + 0.75 (\mathbf{x}_t - \mathbf{x}_{t-1}), \quad (3)$$

and a center of the next field qualifies as the continuation when it lies within $v_{\max}\Delta t$ of both the current position $\mathbf{x}_t$ and the projection $\mathbf{x}_q$, with $v_{\max} = 150 \text{ km h}^{-1}$ and $\Delta t = 6 \text{ h}$. Among the qualifying candidates the one nearest to the projection continues the track. Centers left unmatched at the new time start tracks as genesis events, and tracks left unmatched end as lysis. Only systems poleward of 30° latitude enter the analysis.

Several track-based diagnostics are derived. Track density counts every six-hourly cyclone position, system density counts each cyclone once at its mean position, and genesis density counts the first point of each track. Tracks that merely cross the boundaries of the monthly processing windows would appear as spurious genesis or lysis events, and such truncated events (about 7% of the total) are removed. For every cyclone the lifespan, the maximum depth relative to the outermost closed isobar, the minimum central pressure, the mean cyclone area, and the propagation speed are recorded.

133 *c. Gridded storm coverage*

134 A complementary Eulerian diagnostic measures how long each location spends under cyclone
 135 influence. At every six-hourly time step a binary mask flags the grid cells that lie inside a square
 136 search window of half-width $\sqrt{A}$ grid cells around each cyclone center and whose pressure lies
 137 below the edge pressure of that cyclone,

$$M(\mathbf{x}, t) = 1 \quad \text{where } p(\mathbf{x}, t) < p_e \text{ inside the window,} \quad (4)$$

138 and the masks are regridded to T63 and accumulated into monthly storm days, one quarter of
 139 the flagged six-hourly steps. Storm days at a point therefore scale with the product of cyclone
 140 number and cyclone area footprint. The two factors can carry opposite seasonal cycles, and over
 141 the Southern Ocean the larger size of summer cyclones raises summertime storm days even though
 142 cyclone counts peak in winter. Frequency statements in this paper consequently rest on the track-
 143 based densities, while storm days serve as a smooth field for maps and for measures of cyclone
 144 influence time. The gridded product is available for eleven phases, because the mask product of
 145 the $\omega = 30^\circ$ experiment stems from an inconsistent processing version and is withheld. All figures
 146 use the common set of eleven phases, and the Lagrangian statistics of the withheld phase (which
 147 exist for all twelve) confirm that its omission does not affect any conclusion.

148 An Eulerian measure of synoptic variability provides a further check that involves no tracking
 149 at all. Bandpass sea level pressure variability is computed from the six-hourly fields with the 24-h
 150 difference filter of Wallace et al. (1988), which isolates fluctuations on the synoptic time scales that
 151 define the classical storm tracks (Blackmon 1976), and its monthly standard deviation is analyzed
 152 with the same basins, seasons, and response measures (appendix Fig. A1).

153 *d. Baroclinicity diagnostics*

154 The maximum Eady growth rate serves as the measure of baroclinicity (Eady 1949; Lindzen
 155 and Farrell 1980; Hoskins and Valdes 1990). It is computed as $\sigma = 0.31 |f| |\partial U / \partial z| N^{-1}$ in the
 156 925–700 hPa layer from monthly wind, temperature, and geopotential height. The response to
 157 precession is quantified between the end members p090 and p270, which place perihelion at the
 158 June and December solstices. Because the difference fields are dipolar, plain box averages cancel,

and the response amplitude is instead defined as the root-mean-square of the difference over a basin, normalized by the climatological growth rate of the season. A substitution decomposition in the style of Camargo et al. (2007) splits the difference field into a vertical-shear factor and a static-stability factor, and the share of each factor is obtained by projecting it onto the full difference pattern. The basin averages use the storm-track boxes defined below. This choice matters in the Southern Hemisphere, where a sector beginning at 30°S would admit the seasonal signal of the subtropical winter jet and disguise it as a storm-belt response.

e. Basins and statistics

Three storm-track basins are analyzed, the North Atlantic (30–75°N, 80°W–20°E), the North Pacific (30–75°N, 120°E–120°W), and the Southern Ocean (40–75°S, all longitudes). Basin means are area weighted. The size of the precessional response is reported as the range across the eleven phases, the maximum minus the minimum of the 30-year mean, expressed as a percentage of the phase mean. Uncertainty estimates use the interannual spread of the 30 individual years in each phase, and the significance of the end-member difference maps is assessed with a two-sided Welch t test at each grid point. The local-summer contrasts between extreme phases far exceed the interannual noise, with p values below 10^{-12} in all three basins.

3. Results

a. Winter storm tracks and what storm days measure

The three storm tracks of the ensemble live in winter, exactly as observed. Seasonal-mean genesis counts peak in DJF over the North Atlantic and the North Pacific and in JJA over the Southern Ocean, and system counts follow the same calendar in every phase. Cyclones also cut deepest in the cold season, with winter maximum depths exceeding their summer values by roughly half in the northern basins. The winter identity of the storm season is therefore common to both hemispheres and to all twelve orbital configurations (Table A1).

The gridded storm-day field tells a partly different story, and the difference is instructive. Basin-mean storm days over the Southern Ocean stay high through austral summer (Fig. A6c) although summer cyclones are fewer. The elevation reflects cyclone size rather than cyclone number. Summer systems there are about twenty percent larger in area than their winter counterparts, and

187 storm days count the time a grid cell spends inside any cyclone, so the product of number and
188 footprint hides the winter peak in counts. A diagnostic of cyclone influence time is not a diagnostic
189 of cyclone frequency, and the two must be read together. Frequency statements below therefore
190 rest on the track-based densities, while storm days provide the smooth fields used for maps.

191 *b. The summer paradox*

192 The annual-mean view sets the baseline (Fig. A5). The eleven annual climatologies are close to
193 interchangeable, and the anomalies of each phase from their common mean form weak, smoothly
194 rotating dipoles along the storm tracks. The basin series beneath the maps quantify how little
195 survives in the annual mean, with across-phase ranges of 10%, 17% and 4% in the North Atlantic,
196 the North Pacific and the Southern Ocean, varying smoothly with ω and already hinting at the
197 interhemispheric antiphase, since the northern basins reach their minima near $\omega = 90^\circ$ where the
198 Southern Ocean peaks.

199 Precession barely touches the annual mean but reorganizes the seasons, and the season it se-
200 lects is the quiet one. The seasonal cycles of basin-mean storm days fan out across the phases
201 almost exclusively in local summer (Fig. A6a–c). The across-phase range makes this quantitative
202 (Fig. A6d–f). In the North Pacific the annual range is 17% of the phase mean while the JJA range
203 reaches 67%. The North Atlantic behaves the same way with 10% against 31%. The Southern
204 Ocean repeats the pattern half a year out of phase, with an annual range of only 4% against 16%
205 in DJF. The track-based densities confirm the calendar in every basin, with local-summer ranges
206 of 24%, 43%, and 22% for track density in the North Atlantic, the North Pacific, and the Southern
207 Ocean. The sensitive season is local summer everywhere, and the small annual signal is what
208 survives of a substantial summer swing after the stable winter is averaged in.

209 A tracking-free view points the same way (appendix Fig. A1). Bandpass sea level pressure
210 variability responds most strongly in JJA over the North Pacific, at 36% of its climatology against
211 9% in winter, and winter holds the weakest response of all seasons in both northern basins. This
212 amplitude-weighted metric shifts part of the response within the warm half-year. Over the North
213 Atlantic it peaks in early autumn, matching the secondary September lobe of the forcing itself
214 (Fig. A3c), and over the Southern Ocean its apparent winter share lies on the equatorward flank of

215 the basin box, while poleward of 50°S austral summer leads. The warm-season concentration and
216 the winter quiescence are therefore not artifacts of the tracking algorithm.

217 *c. Orbital anchoring and the interhemispheric mirror*

218 The summer response is anchored to the orbit, not to the calendar. When the basin-mean storm-
219 day anomaly is drawn as a function of calendar month and precession phase, the anomaly ridge
220 climbs through the diagram along the calendar month of perihelion (Fig. A7). The sign is fixed by
221 the orbital geometry. Summers spent near perihelion are quiet and summers spent near aphelion
222 are stormy. North Pacific summer activity reaches its minimum at $\omega = 90^\circ$, the configuration with
223 perihelion at the June solstice, where it falls 32% below the phase mean, and its maximum of
224 +35% near $\omega = 270^\circ$ (Fig. A10a). The North Atlantic swings between -17% and $+15\%$ at the
225 same phases. The Southern Ocean mirrors this behavior, with a DJF maximum at $\omega = 60^\circ$ and a
226 minimum at $\omega = 210^\circ$, because a perihelion that falls in northern summer is an aphelion summer
227 for the south. One lever thus drives the two hemispheres in antiphase across the precession cycle.
228 The extreme-phase contrasts far exceed interannual variability, with Welch p values below 10^{-12}
229 in all three basins.

230 *d. The dynamical bridge runs through summer shear*

231 The seasonal selectivity begins in the forcing itself. The monthly insolation range across the
232 phases at 45°N climbs from below 20 W m^{-2} in midwinter to about 128 W m^{-2} in early summer,
233 and the mirror curve at 55°S peaks near 134 W m^{-2} in December and January (Fig. A3). Perihelion
234 can only amplify a season that already receives sunlight, so the orbital lever presses on summer in
235 each hemisphere and on almost nothing else.

236 Baroclinicity translates the pressure on the seasons into the storm response. The Eady growth
237 rate difference between the end members p090 and p270 is largest in local summer within every
238 storm basin (Fig. A8). Relative to the seasonal climatology the summer response reaches 31%
239 in the North Atlantic, a striking 92% in the North Pacific, and 21% in the Southern Ocean,
240 in each case the seasonal maximum for that basin. A substitution decomposition attributes the
241 pattern almost entirely to the vertical-shear factor, whose projection accounts for 103–109% of the
242 difference field in the three basins, while the static-stability factor contributes near zero. The chain

is short and uniform. Perihelion summers warm the summer hemisphere, flatten the meridional temperature gradient, weaken the thermal-wind shear, and starve the baroclinic growth on which summer cyclones depend. The storm anomalies appear somewhat poleward of the growth-rate anomalies, consistent with the downstream development of baroclinic waves (Chang et al. 2002), and this offset is why pattern measures rather than same-latitude box averages are used throughout. Winter tells a complementary story. The winter growth rate still shifts by 16–18% between the end members, yet the winter storm response stays below 12% in every basin, a point taken up in the discussion.

e. Spatial expression, and what precession actually moves

The end-member difference maps show where the summer signal lives (Fig. A9). In the annual mean the p090 minus p270 difference is weak and patchy. In JJA a pronounced dipole spans the northern basins, with fewer storm days through the midlatitude core and more along the Arctic flank at perihelion summer, the signature of a poleward shift joined to an overall summer weakening. In DJF the significant signal migrates to the Southern Ocean and forms a circumpolar banded structure of opposite sign. About half of the JJA grid points in the northern extratropics pass the 95% significance test, against a far smaller fraction in the annual mean.

The track displacement behind these dipoles can be made explicit (appendix Fig. A2). The storm-day-weighted latitude of the North Pacific summer track migrates from 53°N when perihelion falls in northern winter to 61°N when it falls in northern summer, a poleward excursion of 8° that peaks exactly at $\omega = 90^\circ$, and the track-density-weighted latitude follows the same curve with a 5° swing. The North Atlantic summer track moves 4.4° in the same direction, and the Southern Ocean summer track mirrors the rule with a poleward excursion of 3.8° when perihelion approaches austral summer. A perihelion summer therefore thins the storm population and displaces the survivors poleward, in step with the poleward shift of the baroclinic zone seen in the growth-rate dipole of Fig. A8a. The winter track latitude stays within 1.5–2.8° across the entire cycle, and its small extremes do not align with the solstitial phases, which extends the winter invariance from storm counts to storm position. The shoulder seasons fall in between and carry a telling phase structure, because the autumn excursion peaks near $\omega = 150^\circ$, the configuration that places perihelion in late summer, so the poleward displacement follows the perihelion month through the calendar exactly

272 as the occurrence anomaly does in Fig. A7. The annual-mean track latitude varies by no more than
273 1.3° in any basin.

274 The orbital anchoring of occurrence and position condenses into a single statement (Fig. A11). A
275 first-harmonic fit across the phases gives, for every calendar month, the value of ω at which storm
276 activity reaches its minimum and the track its poleward maximum, and through the sunlit half-year
277 both sequences climb in parallel with the line that places perihelion in that month. The response
278 lags the perihelion calendar by a basin-dependent interval, near one month in the North Pacific,
279 one to two months in the North Atlantic, and slightly over two months in the Southern Ocean,
280 an ordering consistent with the growing oceanic character of the three sectors and with surface
281 thermal inertia setting the delay. Winter months fall off the line, where the response amplitude is
282 too small to anchor. The storm tracks are phase locked to the perihelion calendar wherever the
283 lever can reach them.

284 What precession moves is above all the number and location of cyclones, not their translation
285 speed, and only regionally their depth (Fig. A10b). Local-summer ranges of the frequency and
286 position metrics span 16–67% across the basins. Propagation speed stays within 2–5% everywhere.
287 Cyclone depth separates the basins, and the diagnostics are deliberately kept apart in the manner
288 of Zappa et al. (2013). Over the Southern Ocean the maximum depth varies by only 4% across the
289 phases while frequency varies by 22%, a clean decoupling of frequency from intensity. Over the
290 North Pacific the summer depth range reaches 33% and the lifespan range 23%, so there the whole
291 cyclone population weakens together when perihelion falls in summer, fewer systems that are also
292 shallower and shorter lived. The North Atlantic sits between these regimes. The common element
293 across basins is that occurrence responds most, while the intensity response ranges from absent to
294 substantial depending on how strongly the summer baroclinic reservoir is drained.

295 4. Discussion

296 a. *Why the lever lands in summer*

297 The seasonal selectivity of the storm response is set by orbital geometry before any model physics
298 enters. Precession redistributes insolation in proportion to the insolation a season already has, so
299 the winter hemisphere offers the lever almost nothing to press on (Berger 1978). The monthly
300 forcing amplitude in Fig. A3 makes the floor explicit, with midwinter values near 20 W m^{-2} against

summer values above 125 W m^{-2} at the latitudes of the storm tracks. This part of the result is model independent and would reappear in any climate model, because it restates the geometry that also underlies the summer-energy view of orbital forcing (Huybers 2006). The novelty is not that summer receives the forcing, which the monsoon literature established long ago (Kutzbach and Guetter 1986; Bosmans et al. 2015), but that the summer forcing finds a winter phenomenon and reorganizes it through its quiet season.

b. Cool summers are stormy summers

The mechanism that converts summer insolation into summer storms is the thermal wind. Perihelion summers warm the summer hemisphere most strongly over land and at high latitudes, flatten the meridional temperature gradient, and weaken the vertical shear that feeds baroclinic growth. The factor decomposition confirms that the shear term carries essentially the whole Eady growth rate response in all three basins, while static stability contributes almost nothing. The same chain, with greenhouse warming in place of perihelion, is now observed in the present-day Northern Hemisphere, where amplified summer warming at high latitudes has weakened the summer circulation, the eddy kinetic energy, and the number of strong summer cyclones (Coumou et al. 2015; Chang et al. 2016; Gertler and O’Gorman 2019). Projections extend the analogy, with CMIP-class models thinning summer cyclone activity and the associated baroclinicity through the same shear pathway (Lehmann et al. 2014; Priestley and Catto 2022). The correspondence should be read as a shared mechanism rather than a shared scenario, because the warming trend alters the two hemispheres together whereas precession swings them in antiphase. Under both forcings the rule is the same, and it is the rule in the title. Cool summers are stormy summers.

c. The orbital invariance of the winter storm track

The stability of winter deserves as much attention as the mobility of summer. Winter storm activity varies by less than about a tenth across the entire precession cycle in every basin, and this is only partly a matter of weak winter forcing. The winter Eady growth rate still shifts by 16–18% between the end members, yet winter cyclone numbers barely respond. The winter storm track appears saturated in the sense that ample baroclinicity is available in every phase, so that moderate changes in the growth rate do not translate into changes in occurrence. Whatever its ultimate cause,

the practical consequence is notable. The delivery of winter precipitation to the midlatitude and subpolar continents, the season and pathway that build snowpack and feed ice sheets, is essentially fixed across orbital configurations of the precession cycle. A transient simulation of the last millennium likewise found cyclone counts nearly insensitive to external forcing (Raible et al. 2018; Hess and Chemke 2025), and the present result extends that insensitivity to the far larger insolation excursions of the orbital band, for winter alone.

d. One lever, three basins, and what the records would see

The three basins express one mechanism with different accents, and the diagnostics must be kept apart to see it. Occurrence responds everywhere, with local-summer ranges of 16–67% in frequency and position metrics, while propagation speed is inert. Intensity separates the basins. The Southern Ocean decouples cleanly, with a 4% depth range against a 22% frequency range, the North Pacific weakens as a whole population in perihelion summers, and the North Atlantic lies between. The pattern echoes the long-standing conclusion from anthropogenic scenarios that cyclone number and cyclone intensity respond to forcing through different processes and need not move together (Bengtsson et al. 2009; Ulbrich et al. 2009; Catto et al. 2019). The storm-day diagnostic adds its own caution, because cyclone influence time folds cyclone size into cyclone number, and over the Southern Ocean the seasonal cycle of size alone can invert the apparent storm season. Any proxy that integrates storm influence rather than storm counts inherits this ambiguity.

For the paleoclimate record the seasonal anatomy carries a concrete and testable message. An archive with summer-biased sensitivity in the midlatitude storm belts should carry a pronounced precession signal in storminess, while an annually integrating archive should carry almost none. The seasonal bias of an archive can therefore fabricate or erase an orbital storm signal exactly as it does for temperature (Laepple et al. 2011). The interhemispheric mirror sharpens the prediction, because summer-sensitive storminess records from the two hemispheres should vary in antiphase at the precession period, the extratropical counterpart of the monsoon seesaw (Wang et al. 2008). Existing archives make the test realistic. Precession-band signals are already established in the northern midlatitude westerlies (Li et al. 2019) and in the composition of North Pacific dust, where the precession imprint sits in the moisture-sensitive component while the annually integrating flux follows obliquity (Zhong et al. 2024), and a winter-integrating dust-strength record shows

correspondingly weak precession power (Chen et al. 2014). In the south the westerly belt carries strong 19 and 23 kyr cycles at its subtropical edge (Lamy et al. 2019), with the orbital band recorded at a site depending on its latitude within the belt (Kaiser et al. 2024). The subtropical location of that southern signal is consistent with the present results, because the winter half of the precessional baroclinicity response resides in the subtropical jet equatorward of the storm belt, while the storm-belt response is a summer phenomenon. Records of the southern westerlies also show orbital-band migrations of the belt itself (Lamy et al. 2010; Varma et al. 2012), and glacial-state cyclone modeling has shown how strongly reconstructed storminess depends on which cyclone property a proxy senses (Pinto and Ludwig 2020). A paired-hemisphere test at the precession band is within reach of existing archives.

e. Limitations

Several limitations bound these conclusions. The results rest on one model, one tracking algorithm, and an idealized orbital configuration with eccentricity fixed near its Quaternary maximum, so the simulated amplitudes represent an upper bound that scales with $e \sin \omega$ rather than a reconstruction of any particular epoch. Obliquity is fixed at 24.5° , and interactions between precession and obliquity or ice volume are outside the design. The simulations are equilibrium slices, so transient effects along the precession cycle are not represented. The T63 atmosphere resolves the storm tracks but not mesoscale substructure, and low-level Eady growth rates in the Southern Hemisphere carry a documented sensitivity to boundary-layer stability (Simmonds and Lim 2009). The gridded storm-day product uses eleven of the twelve phases, although the complete Lagrangian statistics indicate that the withheld phase would not alter any conclusion.

5. Conclusions

One rule spans the three storm tracks of this precession ensemble. The orbital lever reaches extratropical cyclones through local summer, the quiet season, while the winter storm track stays essentially fixed across the entire cycle. Summers spent near perihelion are calm and summers near aphelion are stormy, and because one hemisphere's perihelion summer is the other's aphelion summer, midlatitude storminess in the two hemispheres swings in antiphase at the precession period. The signal travels through shear-driven summer baroclinicity and expresses itself chiefly

386 in how many cyclones occur and where they run, with per-cyclone intensity responding only where
387 the summer baroclinic reservoir is drained most deeply. Both expressions are phase locked to
388 the calendar month of perihelion, with a delay of one to two months that grows with the oceanic
389 character of the basin. For the proxy record the lesson is direct. Orbital storminess signals should
390 be sought in summer-sensitive midlatitude archives, and they should appear with opposite sign in
391 the two hemispheres.

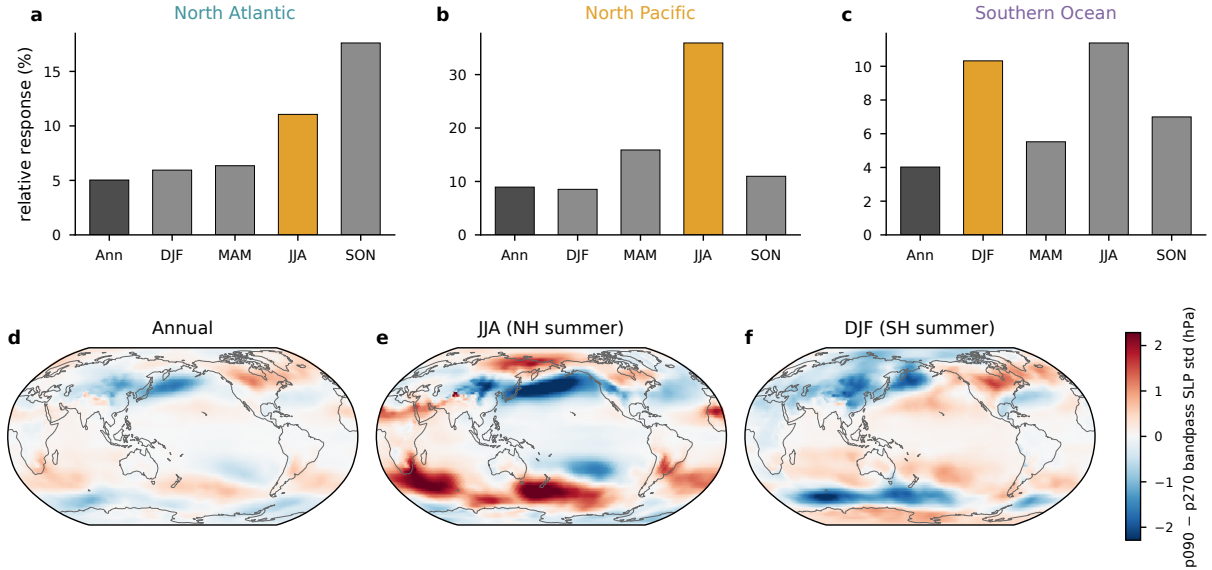


FIG. A1. Tracking-free Eulerian storm-track metric, the monthly standard deviation of 24-h differences of six-hourly sea level pressure. (a)–(c) Relative seasonal response in each basin, the RMS of the p090 minus p270 difference over the storm box as a percentage of the seasonal climatology (as in Fig. A8c), with the local-summer bar highlighted. (d)–(f) The p090 minus p270 difference of the same quantity for the annual mean, JJA, and DJF (hPa).

Acknowledgments. The simulations were performed at the German Climate Computing Center (DKRZ). [TODO funding and grant numbers before submission.]

Data availability statement. The AWI-ESM simulations of the idealized precession ensemble are described in Shi et al. (2025). The cyclone-track database, the gridded storm statistics, and the analysis scripts underlying all figures will be archived and assigned a DOI upon acceptance. [TODO repository DOI.]

APPENDIX

Tracking-independent and positional diagnostics

Figures A1 and A2 present the Eulerian bandpass diagnostic and the seasonal track-latitude analysis discussed in sections 2 and 3.

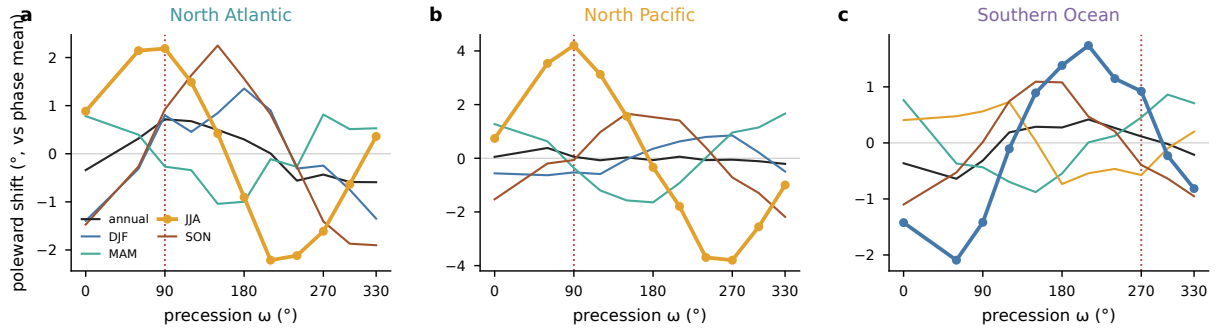


FIG. A2. Storm-track latitude across the precession cycle, the storm-day-weighted mean latitude over each basin sector shown as the poleward anomaly from each season's phase mean, for (a) the North Atlantic, (b) the North Pacific, and (c) the Southern Ocean. Lines give the annual mean and the four seasons, with the local summer of each basin drawn heavy with markers. Vertical dotted lines mark the phase with perihelion in the local summer of each basin.

References

- Bengtsson, L., K. I. Hodges, and N. Keenlyside, 2009: Will extratropical storms intensify in a warmer climate? *Journal of Climate*, **22** (9), 2276–2301, <https://doi.org/10.1175/2008JCLI2678.1>.
- Berger, A. L., 1978: Long-term variations of daily insolation and quaternary climatic changes. *Journal of the Atmospheric Sciences*, **35** (12), 2362–2367, [https://doi.org/10.1175/1520-0469\(1978\)035<2362:LTVODI>2.0.CO;2](https://doi.org/10.1175/1520-0469(1978)035<2362:LTVODI>2.0.CO;2).
- Blackmon, M. L., 1976: A climatological spectral study of the 500 mb geopotential height of the Northern Hemisphere. *Journal of the Atmospheric Sciences*, **33** (8), 1607–1623, [https://doi.org/10.1175/1520-0469\(1976\)033<1607:ACSSOT>2.0.CO;2](https://doi.org/10.1175/1520-0469(1976)033<1607:ACSSOT>2.0.CO;2).
- Bosmans, J. H. C., S. S. Drijfhout, E. Tuenter, F. J. Hilgen, L. J. Lourens, and E. J. Rohling, 2015: Response of the north african summer monsoon to precession and obliquity forcings in the ec-earth gcm. *Climate Dynamics*, **44** (1–2), 279–297, <https://doi.org/10.1007/s00382-014-2260-z>.
- Brayshaw, D. J., B. Hoskins, and E. Black, 2010: Some physical drivers of changes in the winter storm tracks over the North Atlantic and Mediterranean during the Holocene. *Philosophical Transactions of the Royal Society A*, **368** (1931), 5185–5223, <https://doi.org/10.1098/rsta.2010.0180>.
- Camargo, S. J., K. A. Emanuel, and A. H. Sobel, 2007: Use of a genesis potential index to diagnose ENSO effects on tropical cyclone genesis. *Journal of Climate*, **20** (19), 4819–4834, <https://doi.org/10.1175/JCLI4282.1>.
- Catto, J. L., and Coauthors, 2019: The future of midlatitude cyclones. *Current Climate Change Reports*, **5**, 407–420, <https://doi.org/10.1007/s40641-019-00149-4>.
- Chang, E. K. M., S. Lee, and K. L. Swanson, 2002: Storm track dynamics. *Journal of Climate*, **15** (16), 2163–2183, [https://doi.org/10.1175/1520-0442\(2002\)015<02163:STD>2.0.CO;2](https://doi.org/10.1175/1520-0442(2002)015<02163:STD>2.0.CO;2).
- Chang, E. K. M., C.-T. Ma, C. Zheng, and A. M. W. Yau, 2016: Observed and projected decrease of Northern Hemisphere extratropical cyclone activity in summer and its impacts on maximum temperature. *Geophysical Research Letters*, **43** (5), 2200–2208, <https://doi.org/10.1002/2016GL068172>.

- 450 Chen, J., X. Liu, and V. A. Kravchinsky, 2014: Response of the high-resolution Chinese loess grain
451 size record to the 50°N integrated winter insolation during the last 500,000 years. *Geophysical*
452 *Research Letters*, **41** (17), 6113–6120, <https://doi.org/10.1002/2014GL060239>.
- 453 Coumou, D., J. Lehmann, and J. Beckmann, 2015: The weakening summer circulation in the
454 Northern Hemisphere mid-latitudes. *Science*, **348** (6232), 324–327, [https://doi.org/10.1126/](https://doi.org/10.1126/science.1261768)
455 [science.1261768](https://doi.org/10.1126/science.1261768).
- 456 Crawford, A. D., E. A. P. Schreiber, N. Sommer, M. C. Serreze, J. C. Stroeve, and D. G. Barber,
457 2021: Sensitivity of Northern Hemisphere cyclone detection and tracking results to fine spatial
458 and temporal resolution. *Monthly Weather Review*, **149** (8), 2581–2598, [https://doi.org/10.1175/](https://doi.org/10.1175/MWR-D-20-0417.1)
459 [MWR-D-20-0417.1](https://doi.org/10.1175/MWR-D-20-0417.1).
- 460 Danilov, S., D. Sidorenko, Q. Wang, and T. Jung, 2017: The Finite-volume Sea ice–Ocean
461 Model (FESOM2). *Geoscientific Model Development*, **10** (2), 765–789, [https://doi.org/10.5194/](https://doi.org/10.5194/gmd-10-765-2017)
462 [gmd-10-765-2017](https://doi.org/10.5194/gmd-10-765-2017).
- 463 Eady, E. T., 1949: Long waves and cyclone waves. *Tellus*, **1** (3), 33–52, [https://doi.org/10.3402/](https://doi.org/10.3402/tellusa.v1i3.8507)
464 [tellusa.v1i3.8507](https://doi.org/10.3402/tellusa.v1i3.8507).
- 465 Gertler, C. G., and P. A. O’Gorman, 2019: Changing available energy for extratropical cyclones and
466 associated convection in Northern Hemisphere summer. *Proceedings of the National Academy*
467 *of Sciences*, **116** (10), 4105–4110, <https://doi.org/10.1073/pnas.1812312116>.
- 468 Hess, O., and R. Chemke, 2025: Forced changes in the North Atlantic storm track over the
469 last millennium. *Geophysical Research Letters*, **52**, e2025GL115 894, [https://doi.org/10.1029/](https://doi.org/10.1029/2025GL115894)
470 [2025GL115894](https://doi.org/10.1029/2025GL115894).
- 471 Hoskins, B. J., and K. I. Hodges, 2002: New perspectives on the Northern Hemisphere winter
472 storm tracks. *Journal of the Atmospheric Sciences*, **59** (6), 1041–1061, [https://doi.org/10.1175/](https://doi.org/10.1175/1520-0469(2002)059<1041:NPOTNH>2.0.CO;2)
473 [1520-0469\(2002\)059<1041:NPOTNH>2.0.CO;2](https://doi.org/10.1175/1520-0469(2002)059<1041:NPOTNH>2.0.CO;2).
- 474 Hoskins, B. J., and K. I. Hodges, 2005: A new perspective on Southern Hemisphere storm tracks.
475 *Journal of Climate*, **18** (20), 4108–4129, <https://doi.org/10.1175/JCLI3570.1>.

- Hoskins, B. J., and P. J. Valdes, 1990: On the existence of storm-tracks. *Journal of the Atmospheric Sciences*, **47** (15), 1854–1864, [https://doi.org/10.1175/1520-0469\(1990\)047<1854:OTEOST>2.0.CO;2](https://doi.org/10.1175/1520-0469(1990)047<1854:OTEOST>2.0.CO;2).
- Huybers, P., 2006: Early pleistocene glacial cycles and the integrated summer insolation forcing. *Science*, **313** (5786), 508–511, <https://doi.org/10.1126/science.1125249>.
- Kaiser, J., E. Schefuß, J. A. Collins, and Coauthors, 2024: Orbital modulation of subtropical versus subantarctic moisture sources in the southeast Pacific mid-latitudes. *Nature Communications*, **15**, 7512, <https://doi.org/10.1038/s41467-024-51985-4>.
- Kaspar, F., T. Spanghehl, and U. Cubasch, 2007: Northern hemisphere winter storm tracks of the eemian interglacial and the last glacial inception. *Climate of the Past*, **3** (2), 181–192, <https://doi.org/10.5194/cp-3-181-2007>.
- Kutzbach, J. E., and P. J. Guetter, 1986: The influence of changing orbital parameters and surface boundary conditions on climate simulations for the past 18,000 years. *Journal of the Atmospheric Sciences*, **43** (16), 1726–1759, [https://doi.org/10.1175/1520-0469\(1986\)043<1726:TIOCOP>2.0.CO;2](https://doi.org/10.1175/1520-0469(1986)043<1726:TIOCOP>2.0.CO;2).
- Laepple, T., M. Werner, and G. Lohmann, 2011: Synchronicity of antarctic temperatures and local solar insolation on orbital timescales. *Nature*, **471** (7336), 91–94, <https://doi.org/10.1038/nature09825>.
- Lamy, F., R. Kilian, H. W. Arz, J.-P. Francois, J. Kaiser, M. Prange, and T. Steinke, 2010: Holocene changes in the position and intensity of the southern westerly wind belt. *Nature Geoscience*, **3** (10), 695–699, <https://doi.org/10.1038/ngeo959>.
- Lamy, F., and Coauthors, 2019: Precession modulation of the South Pacific westerly wind belt over the past million years. *Proceedings of the National Academy of Sciences*, **116** (47), 23 455–23 460, <https://doi.org/10.1073/pnas.1905847116>.
- Laskar, J., P. Robutel, F. Joutel, M. Gastineau, A. C. M. Correia, and B. Levrard, 2004: A long-term numerical solution for the insolation quantities of the earth. *Astronomy and Astrophysics*, **428** (1), 261–285, <https://doi.org/10.1051/0004-6361:20041335>.

- 503 Lehmann, J., D. Coumou, K. Frieler, A. V. Eliseev, and A. Levermann, 2014: Future changes
504 in extratropical storm tracks and baroclinicity under climate change. *Environmental Research*
505 *Letters*, **9** (8), 084 002, <https://doi.org/10.1088/1748-9326/9/8/084002>.
- 506 Li, Y., Y. Song, Q. Yin, and Coauthors, 2019: Orbital and millennial northern mid-latitude
507 westerlies over the last glacial period. *Climate Dynamics*, **53** (5–6), 3315–3324, [https://doi.org/](https://doi.org/10.1007/s00382-019-04704-5)
508 [10.1007/s00382-019-04704-5](https://doi.org/10.1007/s00382-019-04704-5).
- 509 Lindzen, R. S., and B. Farrell, 1980: A simple approximate result for the maximum growth rate of
510 baroclinic instabilities. *Journal of the Atmospheric Sciences*, **37** (7), 1648–1654, [https://doi.org/](https://doi.org/10.1175/1520-0469(1980)037<1648:ASARFT>2.0.CO;2)
511 [10.1175/1520-0469\(1980\)037<1648:ASARFT>2.0.CO;2](https://doi.org/10.1175/1520-0469(1980)037<1648:ASARFT>2.0.CO;2).
- 512 Merlis, T. M., T. Schneider, S. Bordoni, and I. Eisenman, 2013: Hadley circulation response to
513 orbital precession. part i: Aquaplanets. *Journal of Climate*, **26** (3), 740–753, [https://doi.org/](https://doi.org/10.1175/JCLI-D-11-00716.1)
514 [10.1175/JCLI-D-11-00716.1](https://doi.org/10.1175/JCLI-D-11-00716.1).
- 515 Pinto, J. G., and P. Ludwig, 2020: Extratropical cyclones over the north atlantic and western europe
516 during the last glacial maximum and implications for proxy interpretation. *Climate of the Past*,
517 **16** (2), 611–626, <https://doi.org/10.5194/cp-16-611-2020>.
- 518 Priestley, M. D. K., and J. L. Catto, 2022: Future changes in the extratropical storm tracks and
519 cyclone intensity, wind speed, and structure. *Weather and Climate Dynamics*, **3** (2), 337–360,
520 <https://doi.org/10.5194/wcd-3-337-2022>.
- 521 Raavi, P. H., J.-E. Chu, A. Timmermann, S.-S. Lee, and K. J. E. Walsh, 2023: Moisture control
522 of tropical cyclones in high-resolution simulations of paleoclimate and future climate. *Nature*
523 *Communications*, **14**, 6426, <https://doi.org/10.1038/s41467-023-42033-8>.
- 524 Raddatz, T. J., and Coauthors, 2007: Will the tropical land biosphere dominate the climate–
525 carbon cycle feedback during the twenty-first century? *Climate Dynamics*, **29** (6), 565–574,
526 <https://doi.org/10.1007/s00382-007-0247-8>.
- 527 Raible, C. C., M. Messmer, F. Lehner, T. F. Stocker, and R. Blender, 2018: Extratropical cyclone
528 statistics during the last millennium and the 21st century. *Climate of the Past*, **14** (10), 1499–1514,
529 <https://doi.org/10.5194/cp-14-1499-2018>.

530 Raible, C. C., J. G. Pinto, P. Ludwig, and M. Messmer, 2021: A review of past changes in
 531 extratropical cyclones in the northern hemisphere and what can be learned for the future. *WIREs*
 532 *Climate Change*, **12** (3), e680, <https://doi.org/10.1002/wcc.680>.

533 Shi, X., A. Cauquoin, G. Lohmann, L. Jonkers, Q. Wang, H. Yang, Y. Sun, and M. Werner,
 534 2023: Simulated stable water isotopes during the mid-Holocene and pre-industrial periods
 535 using AWI-ESM-2.1-wiso. *Geoscientific Model Development*, **16**, 5153–5178, [https://doi.org/](https://doi.org/10.5194/gmd-16-5153-2023)
 536 10.5194/gmd-16-5153-2023.

537 Shi, X., M. Werner, H. Yang, Q. Gao, J. Liu, and G. Lohmann, 2025: Precession controls on
 538 climate and water isotope signals in Northern Africa. *Paleoceanography and Paleoclimatology*,
 539 **40**, e2024PA004 999, <https://doi.org/10.1029/2024PA004999>.

540 Sidorenko, D., and Coauthors, 2019: Evaluation of FESOM2.0 coupled to ECHAM6.3: preindus-
 541 trial and HighResMIP simulations. *Journal of Advances in Modeling Earth Systems*, **11** (11),
 542 3794–3815, <https://doi.org/10.1029/2019MS001696>.

543 Simmonds, I., and K. Keay, 2000: Mean southern hemisphere extratropical cyclone behavior in
 544 the 40-year ncep–ncar reanalysis. *Journal of Climate*, **13** (5), 873–885, [https://doi.org/10.1175/](https://doi.org/10.1175/1520-0442(2000)013<0873:MSHECB>2.0.CO;2)
 545 1520-0442(2000)013<0873:MSHECB>2.0.CO;2.

546 Simmonds, I., and E.-P. Lim, 2009: Biases in the calculation of Southern Hemisphere mean
 547 baroclinic eddy growth rate. *Geophysical Research Letters*, **36** (1), L01 707, [https://doi.org/](https://doi.org/10.1029/2008GL036320)
 548 10.1029/2008GL036320.

549 Stevens, B., and Coauthors, 2013: Atmospheric component of the MPI-M Earth System Model:
 550 ECHAM6. *Journal of Advances in Modeling Earth Systems*, **5** (2), 146–172, [https://doi.org/](https://doi.org/10.1002/jame.20015)
 551 10.1002/jame.20015.

552 Ulbrich, U., G. C. Leckebusch, and J. G. Pinto, 2009: Extra-tropical cyclones in the present and fu-
 553 ture climate: a review. *Theoretical and Applied Climatology*, **96** (1–2), 117–131, [https://doi.org/](https://doi.org/10.1007/s00704-008-0083-8)
 554 10.1007/s00704-008-0083-8.

555 Varma, V., and Coauthors, 2012: Holocene evolution of the southern hemisphere westerly winds
 556 in transient simulations with global climate models. *Climate of the Past*, **8** (2), 391–402,
 557 <https://doi.org/10.5194/cp-8-391-2012>.

558 Wallace, J. M., G.-H. Lim, and M. L. Blackmon, 1988: Relationship between cyclone tracks,
559 anticyclone tracks and baroclinic waveguides. *Journal of the Atmospheric Sciences*, **45** (3),
560 439–462, [https://doi.org/10.1175/1520-0469\(1988\)045<0439:RBCTAT>2.0.CO;2](https://doi.org/10.1175/1520-0469(1988)045<0439:RBCTAT>2.0.CO;2).

561 Wang, Y., and Coauthors, 2008: Millennial- and orbital-scale changes in the east asian
562 monsoon over the past 224,000 years. *Nature*, **451** (7182), 1090–1093, [https://doi.org/](https://doi.org/10.1038/nature06692)
563 10.1038/nature06692.

564 Zappa, G., L. C. Shaffrey, K. I. Hodges, P. G. Sansom, and D. B. Stephenson, 2013: A mul-
565 timodel assessment of future projections of North Atlantic and European extratropical cy-
566 clones in the CMIP5 climate models. *Journal of Climate*, **26** (16), 5846–5862, [https://doi.org/](https://doi.org/10.1175/JCLI-D-12-00573.1)
567 10.1175/JCLI-D-12-00573.1.

568 Zhong, Y., and Coauthors, 2024: Orbital controls on North Pacific dust flux during the
569 late Quaternary. *Geophysical Research Letters*, **51**, e2023GL106631, [https://doi.org/10.1029/](https://doi.org/10.1029/2023GL106631)
570 2023GL106631.

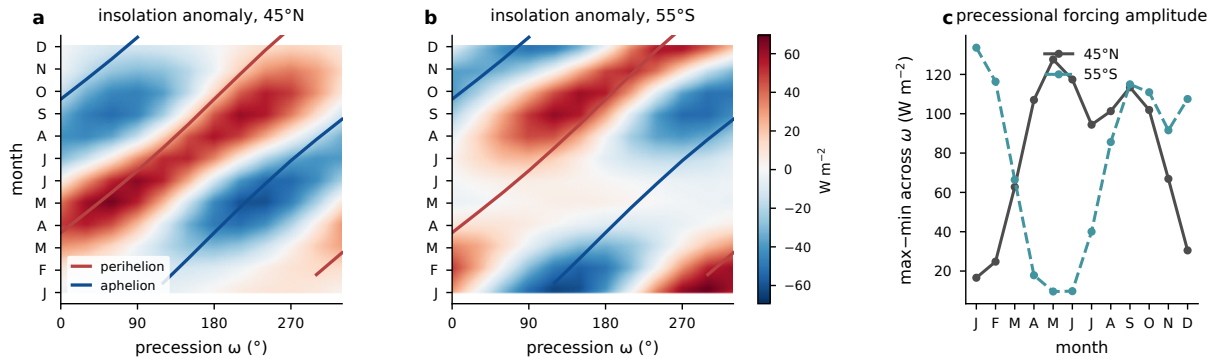
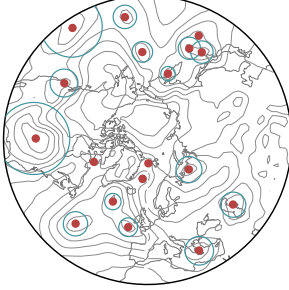
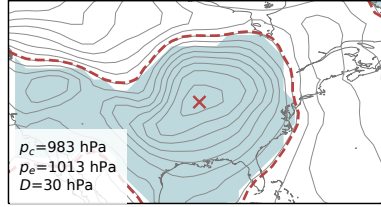


FIG. A3. Precessional insolation forcing from orbital geometry with eccentricity 0.058 and obliquity 24.5° . (a) Monthly insolation anomaly relative to the phase mean at 45°N as a function of calendar month and precession phase ω , with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid. (b) As in (a), but for 55°S . (c) Precessional forcing amplitude by calendar month, the maximum minus minimum monthly insolation across the twelve phases, at both latitudes. Units: W m^{-2} .

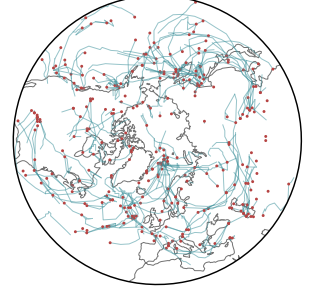
(a) centers + area radii, one 6-h field



(b) outermost closed isobar p_e , mask cells



(c) tracks of one month ($n=407$)



(d)

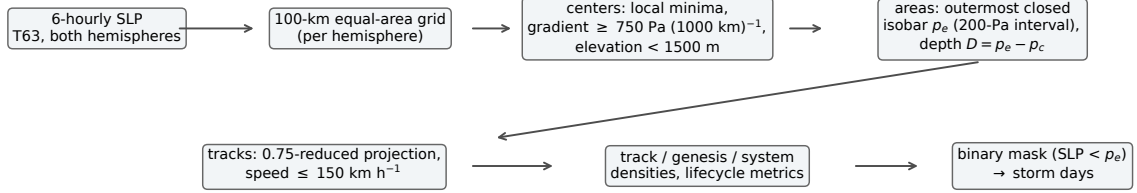


FIG. A4. Cyclone detection and tracking procedure. (a) One six-hourly sea level pressure field of phase p000 (contours, 8-hPa interval) with all detected cyclone centers poleward of 30°N (dots) and their area-equivalent radii $R = \sqrt{A/\pi}$ (circles), 15 January of one model year. (b) The deepest system of that field, with the outermost closed isobar p_e (dashed), the cyclone center (cross), the central and edge pressures and depth annotated, and the binary mask cells (shading) defined by $\text{SLP} < p_e$ inside the $\sqrt{A}$ search window; contour interval 4 hPa. (c) All cyclone tracks of the same month (lines) with genesis points (dots). (d) Processing chain with the key thresholds of Eqs. (1)–(4).

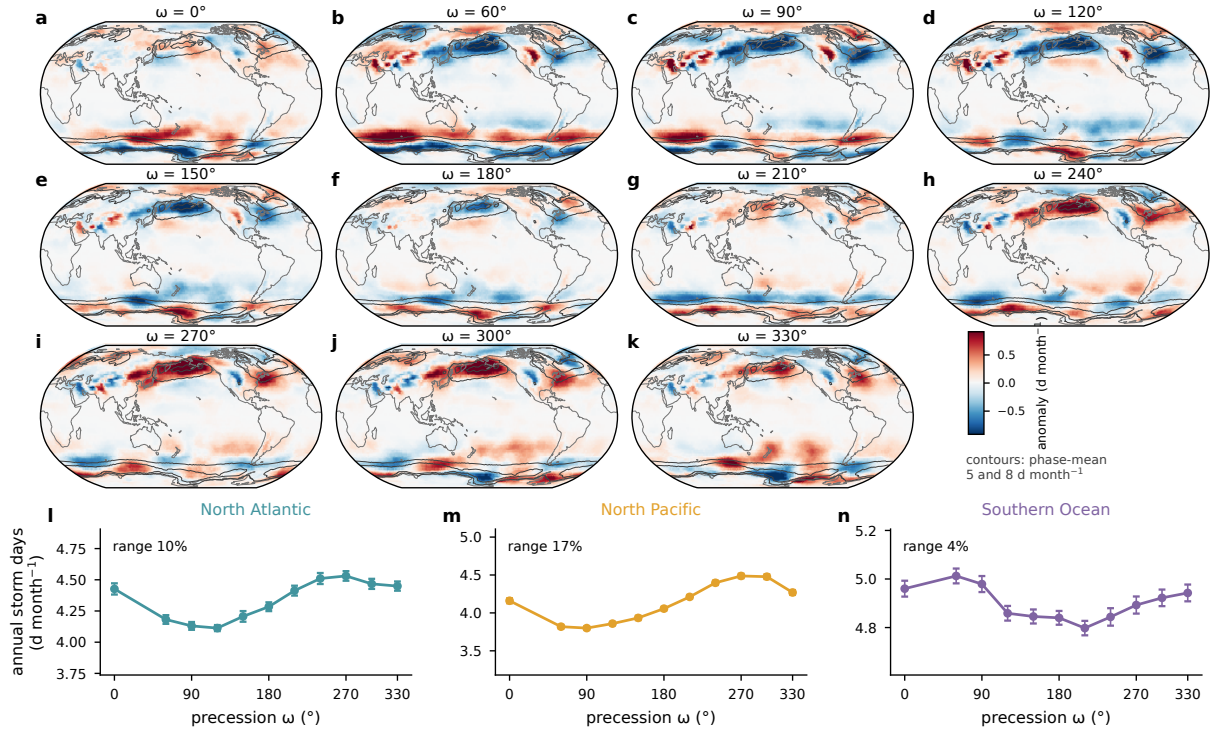


FIG. A5. Annual-mean storm days across the precession cycle. (a)–(k) Annual storm-day anomaly of each phase relative to the eleven-phase mean (shading), with the phase-mean climatology outlined at 5 and 8 d month⁻¹ (contours). (l)–(n) Basin-mean annual storm days against ω with interannual standard errors; the across-phase range as a percentage of the phase mean is annotated in each panel.

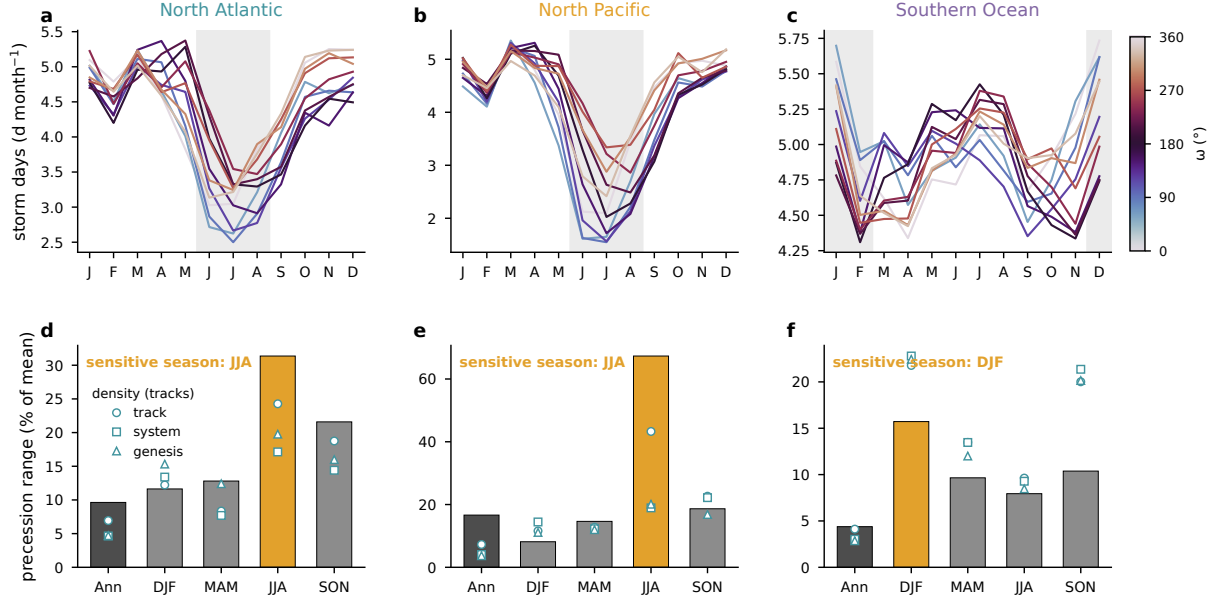


FIG. A6. Seasonal concentration of the precessional storm response. (a)–(c) Basin-mean storm days (d month⁻¹) by calendar month for the North Atlantic, the North Pacific, and the Southern Ocean, one line per phase colored by ω , with local summer shaded. (d)–(f) Across-phase range of storm days (maximum minus minimum, percent of the phase mean) for the annual mean and the four seasons, with the local-summer bar highlighted; open symbols give the corresponding ranges of track, system, and genesis densities from the Lagrangian tracks.

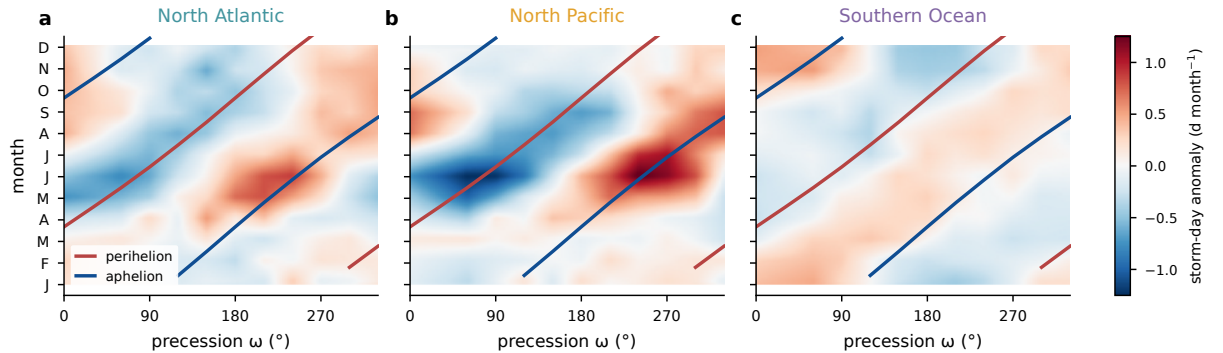


FIG. A7. Basin-mean storm-day anomaly (d month⁻¹) relative to the phase mean as a function of calendar month and precession phase for (a) the North Atlantic, (b) the North Pacific, and (c) the Southern Ocean, with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid.

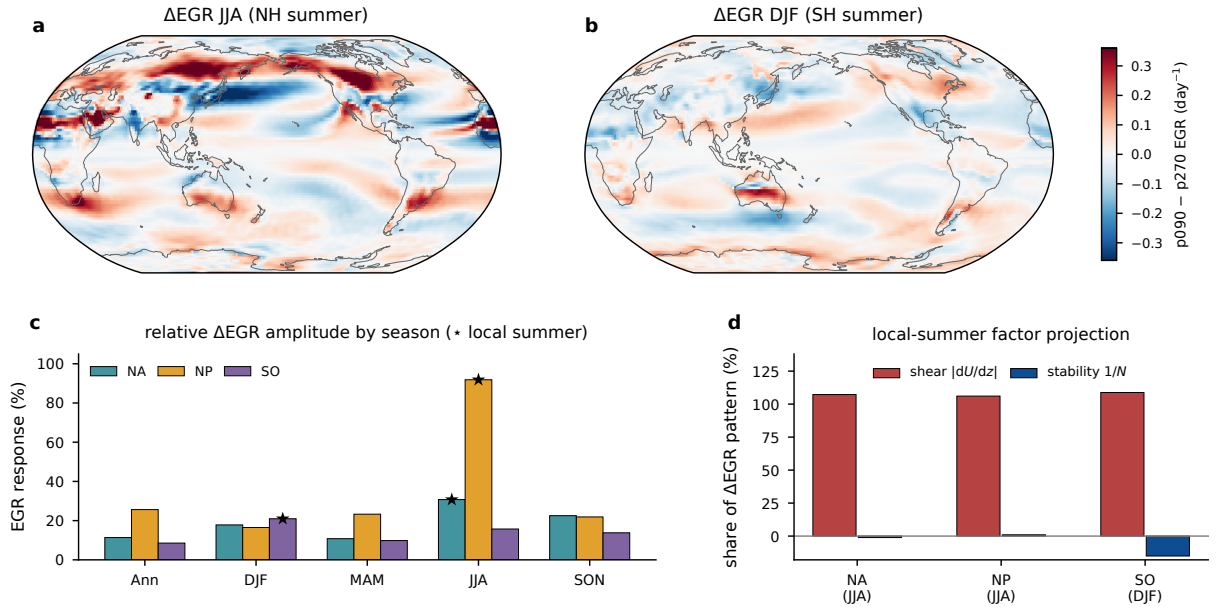


FIG. A8. Precessional response of the maximum Eady growth rate (925–700 hPa). (a) p090 minus p270 difference for JJA and (b) for DJF (day^{-1}). (c) Seasonal response amplitude in each basin, the RMS of the p090 minus p270 difference over the storm box as a percentage of the seasonal climatology, with stars marking local summer. (d) Projection shares of the vertical-shear and static-stability factors in the local-summer difference pattern.

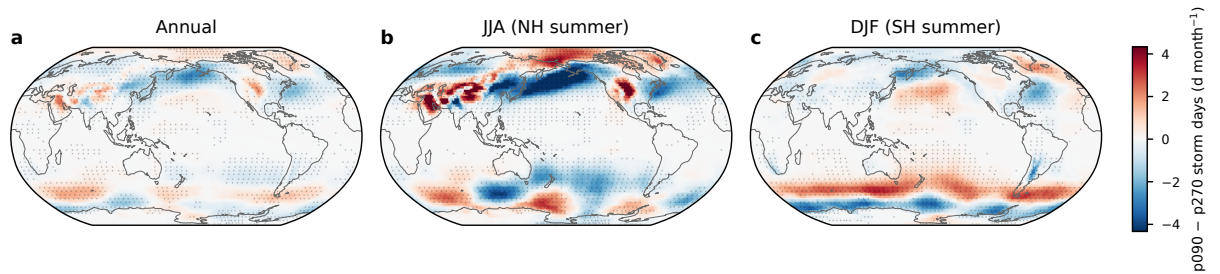


FIG. A9. Difference in storm days (d month^{-1}) between p090 and p270 for (a) the annual mean, (b) JJA, and (c) DJF, on a common color scale. Stippling marks grid points where the difference is significant at the 95% level based on a two-sided Welch t test over the 30 simulated years.

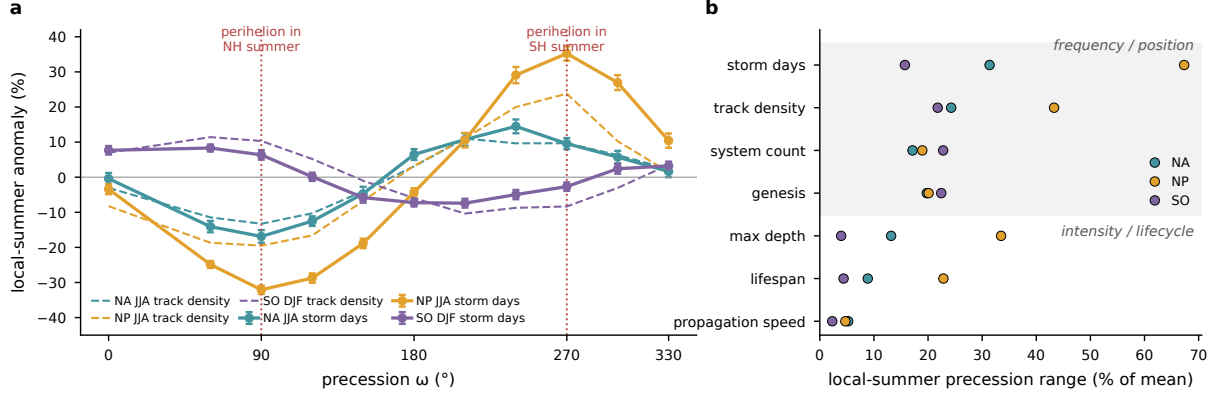


FIG. A10. (a) Local-summer storm activity across the precession cycle, shown as the percent anomaly of basin-mean storm days from the phase mean (solid, with interannual standard errors) and of track density (dashed), for the North Atlantic and the North Pacific in JJA and for the Southern Ocean in DJF; dotted vertical lines mark the phases with perihelion in NH and SH summer. (b) Local-summer across-phase range for frequency and position metrics and for intensity and lifecycle metrics in each basin.

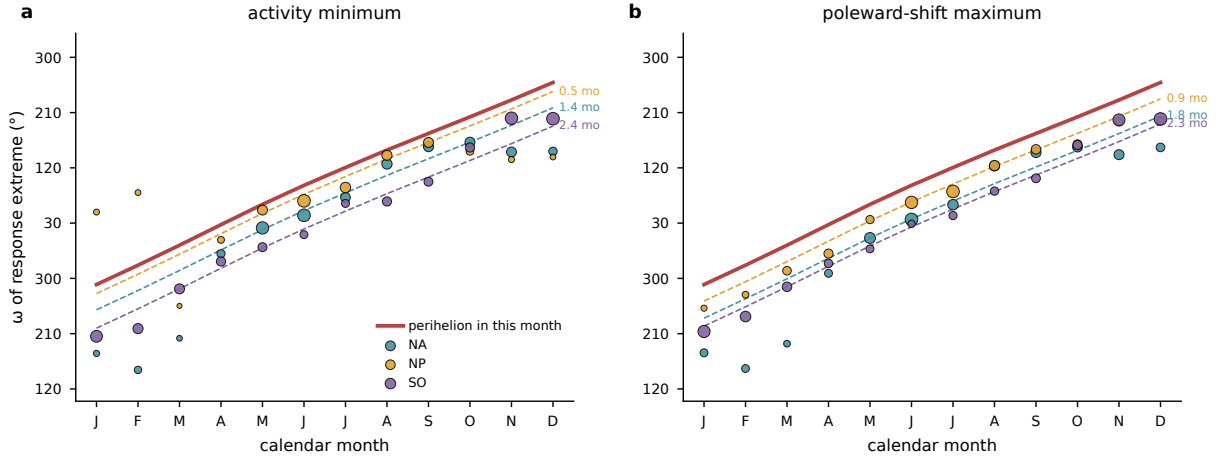


FIG. A11. Phase locking of the storm response to the perihelion calendar. For each calendar month and basin, the phase of the first harmonic across the eleven precession phases of (a) basin-mean storm days, shown as the ω of the activity minimum, and (b) the storm-day-weighted track latitude, shown as the ω of the poleward maximum. Marker size scales with the harmonic amplitude. The red line gives the ω that places perihelion in that calendar month, and dashed lines repeat it delayed by each basin's median lag over the amplitude-strong months (annotated in months).

661 TABLE A1. Three-basin comparison of the precessional storm response. Ranges are max minus min across
662 the 11 phases as a percentage of the phase mean. The storm season is diagnosed from cyclogenesis counts, the
663 sensitive season from the seasonal storm-day range. The EGR response is the RMS p090–p270 difference over
664 the basin sector relative to its climatology; the dominant factor is its projection share.

Basin	Storm season	Sensitive season	Storm-day range ann/summer (%)	Track-density range summer (%)	Summer max/min phase
North Atlantic	DJF (winter)	JJA	10 / 31	24	$\omega=240$ / $\omega=90$
North Pacific	DJF (winter)	JJA	17 / 67	43	$\omega=270$ / $\omega=90$
Southern Ocean	JJA (winter)	DJF	4 / 16	22	$\omega=60$ / $\omega=210$